

Table 4B. Evidence across GENIE3's top-25 candidate regulators

Positi on	Candidate regulator	CRISPRi evidence	Audited literature evidence	Candidate mRNA context	Independent BTLA molecular evidence
1	EGR2	Detected — BTLA-concordant (12/31)	Strong	TCR activation ↑: 1 h, 4 h, 24 h	Not measured in protein or phosphoproteomic assays
2	JUNB ^s	Detected — BTLA-concordant (2/38)	Moderate	TCR activation ↑: 1 h, 24 h	Not measured in protein or phosphoproteomic assays
3	FOXP1	Failed on-target quality control	Moderate	No qualifying transcriptional response	No qualifying association in measured layers
4	HIVEP3	Failed on-target quality control	None	TCR activation ↑: 4 h, 24 h	Not measured in protein or phosphoproteomic assays
5	IRF8 ^s	Detected — BTLA-anti-concordant (6/32)	Weak	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
6	STAT1 ^s	No detected response (0/30)	None	No qualifying transcriptional response	No qualifying association in measured layers
7	BHLHE40	Detected — BTLA-concordant (13/29)	Weak	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
8	PLAGL2 ^s	Not represented in screen	None	No qualifying transcriptional response	No qualifying association in measured layers
9	ZBTB32 ^s	No detected response (0/24)	Weak	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
10	CYCS	No detected response (0/27)	None	No qualifying transcriptional response	No qualifying association in measured layers
11	MDM2	No detected response (0/23)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
12	GAR1	Not represented in screen	None	No qualifying transcriptional response	No qualifying association in measured layers
13	MYC	Detected — BTLA-anti-concordant (1/18)	Moderate	No qualifying transcriptional response	Phosphosite association: 4 h, 24 h
14	ZEB2	Failed on-target quality control	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
15	FOXM1	No detected response (0/10)	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
16	SMAP2	No detected response (0/18)	None	No qualifying transcriptional response	Protein association: 4 h
17	NR4A2	No detected response (0/10)	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
18	CREM ^s	No detected response (0/14)	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
19	ID2	No detected response (0/16)	Weak	TCR activation ↑: 4 h	Not measured in protein or phosphoproteomic assays
20	STAT5B	Detected — BTLA-concordant (1/15)	Weak	No qualifying transcriptional response	Protein association: 20 min
21	TRAF4	No detected response (0/14)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
22	GTF2A2	No detected response (0/14)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
23	TGIF1	No detected response (0/12)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
24	MAF	No detected response (0/11)	Weak	Failed quality control	Not measured in protein or phosphoproteomic assays
25	PRDM1 ^s	Not represented in screen	Strong	TCR activation ↑: 4 h	Not measured in protein or phosphoproteomic assays

S, shared between the GREmLN and GENIE3 top-25 lists.

Candidate regulators were restricted to the pre-specified pySCENIC human transcription-factor/regulator list.

CRISPRi evidence is model-specific because each model can nominate a different target set for the same candidate. The (confirmed/predicted) denominator uses the native predicted-target budget (not a matched top-N budget). Not represented in screen and failed on-target quality control are shown explicitly and are not converted to zero response.

Audited literature evidence uses the Paperclip v2 TCR-inhibition audit (fixed query "<TF> TCR inhibition"). Strong and moderate literature tiers are audited provisional annotations pending documented human verification.

Candidate mRNA context is contextual evidence only (TCR versus unstimulated control). No candidate showed a qualifying incremental BTLA+TCR versus TCR mRNA response; that contrast is therefore omitted from the column.

Independent BTLA molecular evidence is restricted to directly measured qualifying protein abundance or phosphosite change (BTLA+TCR versus TCR) at the source-native thresholds. Inferred regulon activity, kinase activity, BIONIC, co-IP and synapse annotations are not shown. Molecular direction is not interpreted as supportive or opposing because both model rankings are unsigned. Incomplete assay coverage is reported as not measured and is not negative evidence.

Pale green marks detected CRISPRi response, Strong literature, or qualifying independent molecular association. Pale yellow marks qualifying contextual mRNA. Light grey marks not represented, failed QC, not measured, or None.